TEAM 6 / LANI-EV PROJECT

Project Specifications

Engineering Documentation and Performance of a Solar/$LiFePO_4$ Retrofit Kit on a TTU Club Car LSV

- Vehicle restoration
- Solar retrofit
- Performance validation
- Universal mounting kit
- Parked-vehicle solar-tracking concept

TENNESSEE TECHNOLOGICAL UNIVERSITY

Electrical and Computer Engineering Capstone Design

Prepared by	Alec Snodgrass, George Vaporis, Landon Paul, Nathan Galloway, Parker Branson
Faculty advisor	Dr. Charles Van Neste
Project partner	LANI-EV; Jeffery Watters
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Status	Review issue; scope and open design values require confirmation

PROJECT OVERVIEW

Purpose, scope, and priorities

Team 6 will restore and evaluate the inherited Tennessee Tech Club Car low-speed vehicle (LSV), document installation of the LANI-EV solar/$LiFePO_4$ retrofit, measure solar charging versus vehicle discharge, develop a universal solar-panel attachment kit, and design a parked-vehicle solar-tracking control algorithm. The project provides a reproducible engineering record and evidence-based findings for faculty and LANI-EV.

Objective hierarchy

Goal	Responsibility	Priority
1	Characterize and restore the inherited vehicle	Primary / required
2	Implement and document the solar/$LiFePO_4$ retrofit	Primary / required
3	Measure charge-versus-discharge performance	Primary / required
4	Design and prototype a universal solar-panel attachment kit	Primary / required
5	Develop a parked-vehicle solar-tracking algorithm	Primary / concept
6	Quantify environmental impact	Secondary / stretch
7	Quantify economic impact	Secondary / stretch

Completion basis. Goals 1–4 and the Goal 5 concept define required work. Goal 5 hardware deployment is a stretch objective. Goals 6–7 are secondary objectives pursued as stretch work when resources and progress permit; their completion is not a condition of capstone success.

Boundaries and interpretation

- The required prototype is an attachment kit, with a defined compatibility envelope; "universal" does not claim compatibility with every LSV or solar panel.
- Team 1's perovskite research, long-term BMS study, comparative gas-cart program, and revised L2 electrical-improvement project are outside this assignment [S2, S3].
- Multiweek operation without grid charging is a hypothesis, not a guaranteed performance threshold. Report favorable or unfavorable findings with the tested operating conditions.
- Goal 2's roof conversion and Goal 4's attachment kit have a shared interface. The kit does not attach to the retained OEM roof structure; instead replaces the roof entirely.

Requirement conventions

Goal statements describe outcomes. Numbered **G** statements identify sub-goals and their requirements. Each sub-goal has a matching **D** deliverable (for example, G1.1 maps to D1.1). Deliverables may be sections of one shared report rather than separate files; verification evidence demonstrates completion. **TBD** indicates an unresolved value, not a waived requirement. Source-derived requirements and recommended engineering refinements are distinguished in the source disposition.

REQUIRED OBJECTIVE / GOAL 1

Vehicle condition and restoration

Goal 1. Establish the inherited configuration and restore the systems needed for safe retrofit integration and planned testing.

Basis: S1, original Goal 1.

Sub-goals and deliverables

Sub-goal / objective	Deliverable
G1.1 — Record the as-received condition Record identification, property tag, photographs, accessible odometer reading (or unavailability), installed/missing components, disassembly state, and observed defects. Explicitly identify the missing batteries and steering wheel.	**D1.1** Initial vehicle condition report with annotated photographs and component inventory.
G1.2 — Establish the actual configuration Document accessible traction, accessory, charging, and wiring interfaces, component labels, and condition. Identify unknown ratings and unsupported assumptions before selecting components.	**D1.2** Configuration record with an interface diagram, component ratings, and unresolved items.
G1.3 — Plan and document restoration Track steering, braking, chassis, electrical, and other work needed for the intended tests. Assign each defect a corrective action, responsible party, status, and supporting evidence.	**D1.3** Restoration action log with repair/replacement records and verification evidence.
G1.4 — Demonstrate test readiness Before energization or motion testing, record applicable readiness checks, close test-blocking defects, and obtain faculty/sponsor review of remaining limitations. Unresolved unsafe conditions block the affected test.	**D1.4** Readiness checklist identifying completed checks, open restrictions, and reviewer/date.

Baseline policy

A measured lead-acid battery-bank specification sheet is **not required**: the original bank was absent on receipt. Do not describe battery removal, bank weighing, terminal inspection, or pre-conversion operation as work performed when it did not occur.

Historical data may support analysis if its source, applicability, and uncertainty are recorded. Distinguish reference values from measurements. The incomplete inherited cart is not an equivalent fully equipped lead-acid baseline for mass or performance comparisons.

Verification / completion evidence

Inspect the vehicle against D1.1–D1.3 and demonstrate corrected functions relevant to the next work stage. D1.4 shall show closure of required test-enabling work and label unavailable baseline data.

REQUIRED OBJECTIVE / GOAL 2

Retrofit integration and documentation

Goal 2. Install and document the single-unit $LiFePO_4$ battery and solar charging/roof system using a repeatable conversion procedure.

Basis: S1, original Goal 2 and component list.

Sub-goals and deliverables

Sub-goal / objective	Deliverable
G2.1 — Document battery installation Document installation of the single $LiFePO_4$ unit, BMS, tray/adaptor, and connections. Include tools, illustrated sequence, and manufacturer-based checks. Label legacy-bank removal instructions as reference material when not performed.	**D2.1** Illustrated battery conversion manual with installation and retention checks.
G2.2 — Document the electrical system Produce a labeled as-built schematic of the battery/BMS, traction interface, solar controller/panel, wiring, fuses, disconnects, and accessory conversion where needed. Record component identifiers, ratings, polarity, and settings.	**D2.2** As-built wiring schematic and component/settings schedule.
G2.3 — Document solar-roof integration Record OEM roof condition/removal where applicable, the solar-roof installation, final mounting specifications, and component mass changes. Label unavailable baseline mass or reference-based estimates. Coordinate the interface with Goal 4.	**D2.3** Solar-roof installation log with photographs, mounting details, and mass accounting.
G2.4 — Verify system integration Check battery/panel retention, wiring condition/routing, polarity, protection, accessory compatibility, and charging configuration before functional tests. Use selected battery/BMS and controller documentation to establish settings.	**D2.4** Integration checklist, functional-check results, and deviation/corrective-action record.

Required system elements

- One $LiFePO_4$ battery, integrated or separate BMS, and secure tray/adaptor.
- Solar-integrated roof/panel and compatible charge controller; MPPT/PWM selection TBD.
- Appropriate cables, connectors, fuses, disconnects, and accessory voltage conversion if required.

Verification / completion evidence

Review D2.1–D2.4 against the installed configuration. Physical inspection and recorded functional checks shall demonstrate that the installation agrees with the documentation and applicable component limits.

REQUIRED OBJECTIVE / GOAL 3

Charge-versus-discharge analysis

Goal 3. Measure solar charging and vehicle discharge under reproducible conditions and determine the observed post-conversion energy balance.

Basis: S1, original Goal 3.

Sub-goals and deliverables

Sub-goal / objective	Deliverable
G3.1 — Define the test protocol Define speed, payload, terrain/route, duration, initial SOC, environment, equipment, sampling, repetitions, safety limits, comparison conditions, and data-quality criteria before testing.	**D3.1** Reviewed performance-test protocol and instrumentation plan.
G3.2 — Measure charge and discharge Execute the protocol. Record time-aligned solar charge current, relevant voltages, sunlight conditions, load, distances where applicable, and interruptions. Define sensor locations and instruments used.	**D3.2** Raw time-series data, test/configuration log, and instrumentation record.
G3.3 — Analyze and report performance Provide processed data, charge/discharge plots, and an energy balance. State uncertainty, missing data, and limitations; determine whether the results support sustained solar-assisted operation under the tested conditions.	**D3.3** Analysis report with processed tables, plots, calculation files, and findings.

Measurement and analysis basis

- Declare the measurement boundaries. Currents at different voltages cannot be compared directly as an energy balance; compare power and integrated energy.
- Calculate $P(t) = V(t)I(t)$ and integrate over the stated interval. Explain treatment of auxiliary loads and conversion losses.
- Prefer irradiance in W/m^2. Foot-candles measure illuminance; label them accordingly and do not equate them to irradiance without a justified conversion.
- Net battery current alone does not independently measure solar input and traction discharge. Resolve instrumentation adequacy before claiming G3.2 complete.

Verification / completion evidence

Check D3.2–D3.3 against D3.1 and reproduce representative calculations from raw data. A complete, defensible evaluation satisfies this objective; favorable results are not required.

No minimum range, recharge rate, solar energy fraction, or grid-free operating duration is established. Optional testing of the Goal 5 tracking concept is separate from the required fixed-configuration analysis.

REQUIRED OBJECTIVE / GOAL 4

Universal solar-panel attachment kit

Goal 4. Design, model, fabricate, and document an adaptable, non-invasive kit that securely and aesthetically fits a solar panel to multiple major-brand LSVs.

Basis: S2, original Goal 3. S3 describes a different Team 1 task.

Sub-goals and deliverables

Sub-goal / objective	Deliverable
G4.1 — Survey three major brands Survey three major LSV brands. Record selected models, roof geometry/dimensions, materials, attachment points, clearances, and data sources. EZGO, Yamaha, and Club Car are source examples; selections require confirmation.	**D4.1** Three-brand LSV roof specification matrix.
G4.2 — Design for at least two brands Develop a common kit architecture adaptable to selected models from at least two brands. Identify shared parts, model-specific adaptors, adjustment ranges, and unsupported configurations.	**D4.2** CAD concept, preliminary parts list, and compatibility matrix.
G4.3 — Define attachment and stability Prioritize non-invasive installation, retention, aerodynamic stability, and exterior fit. Assess modifications, load paths, clearances, loosening/vibration, surface protection, and exposure.	**D4.3** Design-basis record with attachment details, load cases, calculations, and acceptance criteria.
G4.4 — Fabricate the prototype Fabricate a functional kit using suitable materials. Record as-built materials, fasteners, dimensions, and deviations from CAD. A functional attachment-kit prototype remains required.	**D4.4** Physical prototype and as-built fabrication record.
G4.5 — Verify compatibility and stability Document fit evidence for at least two brands using reviewed measurements, CAD/interface checks, and prototype fit checks. Demonstrate installation on a vehicle or representative fixture.	**D4.5** Fit/installation and structural/stability verification report against the resolved criteria.
G4.6 — Release the engineering package Release editable CAD, dimensioned drawings, BOM, and illustrated instructions covering adjustments, model-specific parts, tools, fasteners, checks, and compatibility limits.	**D4.6** Final engineering drawings, CAD, bill of materials, and installation guide.

Verification / completion evidence

D4.1 shall cover three brands; D4.2 and D4.5 shall support at least two. D4.4 shall establish a functional prototype, and D4.6 shall match its released configuration. No numerical load, mass, cost, or installation-time limit is supplied by the original sources.

Goal 5 interface. Consider clearance and mounting needs for a future tilting panel. Powered tilt is not required for Goal 4 completion; a full-scale moving roof would require separate mechanical/load validation.

REQUIRED OBJECTIVE / GOAL 5

Parked-vehicle solar-tracking concept

Goal 5. Develop and evaluate a control algorithm that uses vehicle-state, compass, clock, and configured location inputs to command four servos for limited panel tilt toward the sun while parked.

Basis: New Team 6 design concept; technical context E1–E2.

Sub-goals and deliverables

Sub-goal / objective	Deliverable
G5.1 — Define inputs and design boundaries Define the transmission/vehicle-state signal, heading convention, date/time/time-zone data, configured latitude/longitude, and four servo outputs. State the assumed panel/linkage geometry, level-parking assumption, and tilt limits.	**D5.1** Concept block diagram and input/output specification with assumptions and TBD parameters.
G5.2 — Calculate the desired orientation Select a documented solar-position method using date, time, and location. Transform the calculated sun direction into the cart reference frame using compass heading; constrain target tilt to the assumed mechanism limits.	**D5.2** Sun-position and orientation algorithm, equations/flowchart, and representative calculated cases.
G5.3 — Coordinate four servo commands Map the target orientation to four coordinated servo commands using a declared linkage model. Include calibration offsets, travel/rate limits, and constraints that prevent incompatible commands to one rigid panel.	**D5.3** Parameterized command-generation software and servo/linkage mapping; assumed geometry clearly labeled.
G5.4 — Define parked and fault behavior Permit tracking only with a valid parked-enable condition. Define initialization, tracking, stow/disabled, and fault states; inhibit tracking on non-parked or invalid inputs. Define a bounded stow sequence and distinguish commanded stow from confirmed physical position.	**D5.4** State diagram, transition table, and implemented enable/fault/stow logic.
G5.5 — Verify the software concept Run repeatable simulations with representative times, dates, locations, and headings, plus state changes and invalid inputs. Check solar calculations against a documented reference and check all four outputs against the assumed motion limits.	**D5.5** Runnable simulation/test harness, expected-versus-actual results, and command plots.
G5.6 — Document feasibility and handover Assess limitations, calibration, servo/driver/power needs, mechanical compatibility with Goal 4, and work needed for deployment. Record any bench demonstration separately from required software evidence.	**D5.6** Design report, editable code, run instructions, and implementation recommendations.

Scope limit. Required work is the algorithm and simulation. A four-servo bench demonstration or full vehicle installation is optional. Servo torque, linkage geometry, tilt range, accuracy, and energy gain are not established. This is mechanical sun tracking; electrical MPPT adjusts the PV operating point and is not a required new controller design [E1–E2].

SECONDARY OBJECTIVES / GOALS 6 AND 7

Environmental and economic stretch work

These objectives retain the original Team 6 environmental and economic scope, renumbered to accommodate the required mounting kit and new tracking concept. Pursue them only when schedule and resources permit.

Goal 6 — Quantify environmental impact

Sub-goal / objective	Deliverable
G6.1 — Assess a single vehicle Estimate grid-electricity and battery-material impacts using an explicit baseline, period, and measured or sourced inputs. Distinguish actual recycling/disposal from hypothetical landfill diversion; absent batteries cannot be counted as measured Team 6 waste diversion.	**D6.1** Optional single-unit environmental assessment with calculations, assumptions, and sources.
G6.2 — Model a state-level fleet Extrapolate supported unit findings using declared fleet size, duty cycle, grid-emissions factors, and uncertainty. Distinguish projected carbon and utility-cost effects from measured savings.	**D6.2** Optional fleet environmental scenario model and summary.

Goal 7 — Quantify economic impact

Sub-goal / objective	Deliverable
G7.1 — Assess single-vehicle economics Calculate conversion cost, operating-cost differences, ROI, and payback for a defined period. Include applicable electricity, maintenance, and labor; fuel savings require an explicit fuel-powered comparison.	**D7.1** Optional single-unit ROI/payback report with assumptions and calculation files.
G7.2 — Model fleet economics Extrapolate the unit model to a hypothetical state-level fleet. Report cumulative net benefit and sensitivity to major assumptions.	**D7.2** Optional fleet economic extrapolation model and summary.

Verification / completion evidence

If performed, calculations shall be reproducible from declared inputs. Fleet projections shall trace to the corresponding unit-level model and distinguish measured values from assumptions.

Goals 6–7 are secondary stretch objectives. Neither completion nor a positive environmental/economic result is required for project acceptance.

VERIFICATION AND HANDOVER

Required deliverables and completion criteria

Each D identifier corresponds directly to its G sub-goal. A single report may contain several deliverables if sections and evidence are clearly indexed.

Deliverables	Verification / completion evidence
D1.1–D1.4	Condition inventory, actual interfaces, restoration actions, and readiness records agree with the vehicle; test-blocking defects are closed.
D2.1–D2.4	Conversion manual, wiring/settings record, roof log, and checks match the installed and demonstrated configuration.
D3.1–D3.3	Reviewed protocol, raw logs, and reproducible analysis establish charge/discharge behavior with limitations documented.
D4.1–D4.6	Three-brand survey, at least two-brand design compatibility, functional kit prototype, fit/stability evidence, and released installation package.
D5.1–D5.6	Input definitions, orientation/servo algorithms, operating-state logic, runnable simulation, test results, and implementation handover. Physical tracker deployment is optional.
DH	Consolidated status matrix listing every required G sub-goal, corresponding D evidence, reviewer/date, status, and unresolved limitations.

Goal 5 minimum software verification cases

- Valid parked input, configured location, date/time, and heading produce bounded commands for all four servos under the declared geometry.
- Representative dates/times and compass wraparound exercise the sun-position and cart-frame transformation; compare against a documented reference within a tolerance defined before testing (T09).
- Non-parked/unknown vehicle state, invalid or stale clock/heading data, restart, and below-horizon sun invoke the declared disabled, stow, or fault response without continuing normal tracking.
- Requested tilt outside the modeled envelope is limited; coordinated commands respect travel/rate and linkage constraints.

Deployment boundary. A simulated parked input or stow command does not prove that the vehicle is stationary or the panel is physically stowed. Before physical use, define actual signal semantics, feedback needs, motion protection, and mechanical clearance. Neutral alone shall not be assumed to prove a parked vehicle.

Review sequence and status

Confirm restoration readiness before affected integration/tests; resolve Goal 4 criteria before fabrication; review the Goal 3 protocol before data collection; and freeze Goal 5 assumptions before software verification. Assemble DH at handover.

Status terms: not started; in progress; verified; not met; or deferred by recorded scope change. An unresolved required item remains open. Goals 1–4 and the Goal 5 concept/software deliverables define required work; Goals 6–7 and tracker hardware demonstration remain optional.

DOCUMENT CONTROL

Source reconciliation and revision record

Source register

S1 `0. Team6_Original Project Specs.pdf`, 4 pages; undated. Original Team 6 title, vehicle identifier, goals, and equipment list.

S2 `Team1_Original Project Specs.pdf`, 9 pages; marked draft. Mounting goal: pp. 3–4; illustrative equipment appendix: pp. 6–9. January 16, 2026 is a proposed start date, not a revision date.

S3 `Team1_Updated Project Specs.pdf`, 4 pages; undated. Updated status follows the filename. Goal 3 (p. 3) concerns an L2 electrical improvement.

Disposition of differences

Issue	Resolution
Vehicle baseline	This document supersedes S1's failed battery bank description. Goal 1 includes restoration; the measured legacy-bank sheet is removed.
Goal numbering	S1 Goals 1–3 retained; S2 mounting work becomes Goal 4; this revision adds Goal 5; original environmental/economic goals become optional Goals 6–7.
Updated Team 1 task	S3 replaces its mounting task with an L2 electrical improvement. Its conditional prototype and spring deadline do not apply to Goal 4.
Roof architecture	S1 describes roof replacement; S2 describes roof attachment. Resolve their shared interface, including future tracker clearance.
Example values	S2 examples (36/48 V, 100 Ah, 100–410 W, 58-inch mounts, M6 fasteners) are not mandatory specifications. US-made procurement is a preference.
Engineering refinements	G1.3–G1.4, G2.4, G4.5, measurement controls, and verification criteria formalize the scope. Goal 5 details the development of the tracker into software plus conceptual hardware; no numerical performance limit is invented.

Technical context for Goal 5

E1: NREL, Solar Position and Intensity (SOLPOS): solar position from date, time, and location. E2: Texas Instruments, TIDA-01556: electrical MPPT implementation. These references clarify the concept; no hardware or performance values are adopted.

Revision history

Rev.	Issue date	Description
0.1	2026-09-27	Synthesized restoration, conversion, charging analysis, and mounting scope; made environmental/economic work optional.
0.2	2026-09-27	Removed title numbers; paired G sub-goals with D deliverables; added tracking Goal 5; renumbered stretch Goals 6–7.